MT Classes, Motihari

Reproduction: How Life Continues

Class 9 Science, Chapter 11 (NCERT Exploration, new syllabus)



Budding, spores and grafting, meiosis and variation, flowers, pollination and fertilisation, and human reproduction, explained with the textbook's own diagrams, six hands-on tools, every NCERT question solved and a quiz.

Start What is reproduction?

Every living thing is born, grows, matures, **reproduces** and finally dies. A mango tree grows old and dies, but its seeds become new mango trees. Cows give birth to calves, dogs to puppies, humans to children. This is how life on Earth goes on.

Reproduction

The biological process by which living beings **produce new individuals of their own kind**. It keeps the species going and passes genetic information to the next generation.

Two ways to reproduce

Asexual	Sexual
One parent No gametes, no fertilisation.	**Two parents** Male and female gametes fuse (fertilisation).
Offspring are copies Genetically identical to the parent (clones).	**Offspring show variation** A new mix of characters from both parents.
Cell division: mitosis Fast; good when conditions are favourable.	**Cell division: meiosis** Makes gametes with half the chromosomes.
Examples	**Examples**

Bryophyllum, potato, yeast, hydra, bread mould.

Flowering plants, frogs, birds, humans.

Why do small differences matter?

In sexual reproduction, children get a **mix** of their parents' characters, so they differ a little from their parents. Over many generations, such differences help living beings **adapt** to changing environments, and sometimes even give rise to new kinds of species.

Think It Over

Try to answer first, then tap to check.

– **When does a farmer prefer asexual or sexual methods of reproduction for crop production?**

Asexual (vegetative) methods when the farmer wants many plants that are **exactly like a good parent plant**, quickly: e.g. sugarcane and money plant by cutting, potato and ginger from underground stems, mango and rose by grafting, banana by tissue culture. The fruit quality stays the same and plants are ready faster.
Sexual methods (seeds) when the farmer wants **new, improved varieties** (high yield, disease resistance) through hybridisation, or for crops that are naturally grown from seed such as wheat, rice, maize and pulses.

– **Why do most complex animals and flowering plants use sexual reproduction, while simple organisms like yeast and hydra mainly reproduce asexually?**

Complex organisms live long, have few offspring and face changing environments. Sexual reproduction gives **variation**, so some offspring can survive new conditions and the species can adapt and evolve. Simple organisms like yeast and hydra can multiply very **fast** by mitosis (budding) when food and warmth are plentiful; one parent is enough and no energy is spent finding a mate. Speed matters more to them than variation.

11.1 Asexual reproduction

Asexual reproduction is seen in many **unicellular organisms** (bacteria, amoeba, yeast), in **simple multicellular animals** (hydra, sponge) and in many **plants**.

Vegetative propagation

Many plants grow new shoots and roots from their existing parts, **without seeds**:

- **Potato, ginger**: fleshy underground stems sprout new plants.
- **Money plant, sugarcane**: stem cuttings grow into new plants.
- ***Bryophyllum***: tiny plantlets sprout along the edge (notches) of the leaf and grow into new plants (Fig. 11.1).

New plants arising from the **vegetative parts** (root, stem, leaf: the growing parts, not flowers or seeds) is called **vegetative propagation**.



Fig. 11.1 *Bryophyllum* leaf sprouts tiny plantlets

Key point

Vegetative propagation involves **only one parent**, so it produces **genetically identical** individuals.

11.1.1

How is vegetative propagation helpful in agriculture?

Because the new plants are exact copies of the parent, scientists and gardeners (horticulturists) use **cutting, grafting, layering** and **tissue culture** to grow large numbers of plants with the **same desirable qualities** (sweet fruit, big flowers, high yield). Activity 11.1 asks you to watch gardeners or farmers doing these.

1. Cutting

1. Cut the overgrown branches at the end of the growing season and prepare cuttings; note their length and count the **nodes and internodes**.
2. Collect the shoot cuttings in the **morning**.
3. Remove the leaves from the lower half of each cutting.
4. Push each cutting about **half its length** into soil mixed with compost, at an angle of about **45–60°**.
5. Water regularly. Roots grow from the buried nodes and a new plant forms.



Fig. 11.2 Cutting: a stem piece planted at an angle

Used for: rose, sugarcane, money plant, hibiscus, grapes, bougainvillea.

2. Grafting

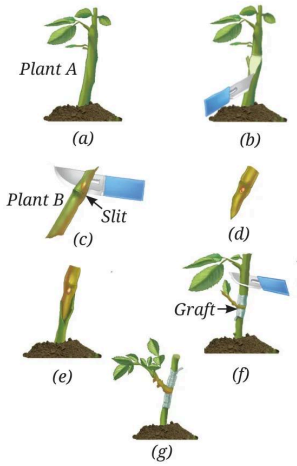


Fig. 11.3 Steps of grafting (a to g)

1. Take a healthy **rooted** plant, **Plant A** (e.g. wild rose). This part with roots is called the **stock**.
2. Take a healthy stem piece of another variety, **Plant B** (e.g. yellow or pink rose). This is the **scion**.
3. Make a **slit** (wound) on a twig of Plant A.
4. Insert and fit the stem of Plant B into the slit.
5. Wrap the joint with cotton cloth or film so pests cannot enter until it heals. Cut the other branches of Plant A.
6. Water regularly. Plant B grows using the roots of Plant A.

Remember: stock and scion

Stock = **S**oil side (has the roots). **Sc**ion = the **Sc**ene on top (gives the flowers or fruit we want). The new plant gives the fruit or flowers of the **scion** with the strong roots of the **stock**. Mango, guava, apple and rose are often grafted.

3. Layering

1. Bend a flexible, thin twig (e.g. lemon) and **bury its middle part** under the soil, still attached to the parent.
2. Water regularly.
3. In **10–15 days**, roots develop from the buried part.
4. Now cut the twig from the parent plant. It grows as a new plant.



Fig. 11.4 Layering

Used for: lemon, jasmine, grapes, strawberry.

4. Tissue culture

A few cells from the **shoot tip (apical meristem)** are grown on a nutrient medium in a sterile glass jar. They form a mass that grows into thousands of tiny **plantlets**, which are then planted in fields.

It has revolutionised **banana farming**: farmers get mass-produced, **virus-free**, healthy plantlets that give high yields. This is also asexual reproduction.

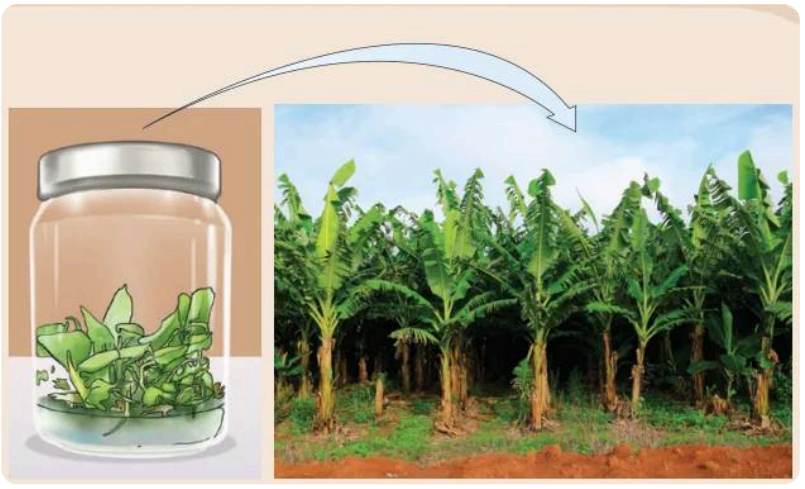


Fig. 11.5 Plant tissue culture

Compare the four methods

Method	What is done	Example	Main benefit
Cutting	Stem piece planted in soil	Rose, sugarcane	Simple, cheap
Grafting	Stem of one plant joined onto a rooted plant	Mango, rose	Combine good fruit with strong roots
Layering	Twig buried while still attached to parent	Lemon, jasmine	Parent feeds the twig until it roots
Tissue culture	Cells grown in a lab on nutrient medium	Banana, orchids	Thousands of disease-free plants fast

Bridging Science and Society: KVKs

Krishi Vigyan Kendras (KVKs) of the **Indian Council of Agricultural Research (ICAR)** train farmers in modern grafting so they can grow high-yield fruits, and teach them marketing and government schemes to boost their income.



Budding and spore formation

Budding in yeast (Activity 11.2)

1. Take 20 mL sugar solution (1 g in 10 mL) in a test tube, add a pinch of yeast and plug with cotton.
2. Keep it in a warm place for 1–2 hours.
3. Put a drop on a slide, cover with a coverslip and see under a microscope.

You see small round **outgrowths (buds)** on the parent yeast cells. Yes, the yeast is **duplicating**: each bud becomes a new yeast cell. Often buds stay attached and form a short chain.

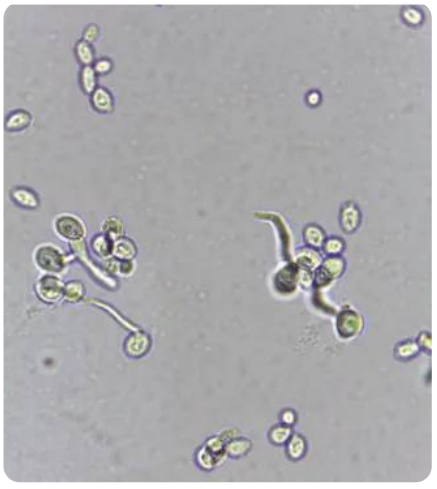


Fig. 11.6 Yeast with outgrowths (buds)

Budding in hydra



Fig. 11.7 Budding in hydra

Repeated cell division at **one spot** on the parent

↓

A small outgrowth: the **bud**

↓

Bud grows tentacles and a mouth

↓

Bud and lives on its own

This is called **budding**. A hydra can carry several buds at the same time.

Where do microbes come from? Louis Pasteur

People once thought living things arose **spontaneously** from non-living matter (like mould "appearing" on food). **Louis Pasteur**'s experiments proved that **new life always comes from pre-existing life**. His work gave the **germ theory of disease**, strengthened the **cell theory** (all cells come from pre-existing cells) and led to sterilisation of food and instruments.

Spore formation in bread mould (Activity 11.3)

1. Moisten a slice of bread or roti and place it on wet cotton covered with tissue paper in a closed box (a **moist chamber**).
2. Keep it in a warm, dark place (25–35 °C). Keep it moist and observe daily without touching.
3. After about 3 days, fuzzy **mould** grows. Stain a little with cotton blue and see it under a microscope.

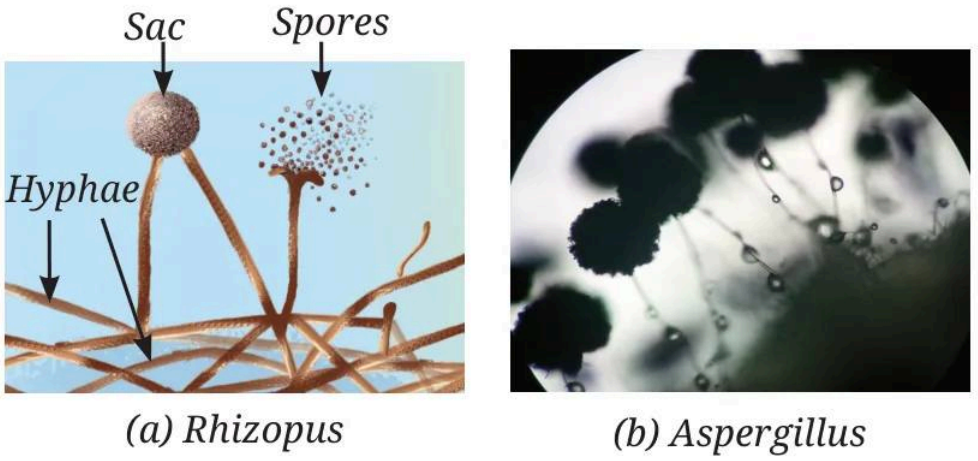


Fig. 11.8 Fungi: (a) *Rhizopus* (spores in a sac), (b) *Aspergillus* (spores on a swollen vesicle)

Hyphae
Thin, thread-like structures that spread over the bread and absorb food.

Sporangium (sac)
Round sac at the tip of an upright hypha. Spores form inside it (*Rhizopus*).

Spores
Tiny, light, usually single-celled. Millions from one colony; float in air.

The mould was not on the fresh bread. It grew from **spores already in the air** that settled on the moist bread. Spores wait for **moisture, warmth and nutrients**, then germinate quickly into a new mould.

Why we refrigerate food

Spores need warmth and moisture. **Low temperature slows or stops** their growth. Before refrigerators became common (about 100 years ago), fresh food lasted only 1–2 days. Refrigeration and deep freezing made fruits, vegetables and dairy available all year and prevented spoilage by moulds and bacteria.

Fungi are useful too

Fungi break down organic waste and pollutants, help remove **heavy metals** from industrial waste, and give us **antibiotics** such as **penicillin** (and the related amoxicillin), which have saved countless lives. Some fungi (for example *Aspergillus* and *Pestalotiopsis* species) have been found to slowly break down certain plastics.

The common thread: mitosis

Mitosis → identical copies (clones)

Mitosis is cell division that makes **two daughter cells with the same number of chromosomes, identical to the parent cell**. All asexual reproduction is based on it, so the offspring are **clones**. It is **fast** and helps a population grow quickly when conditions are good.

Asexual methods at a glance

Method	How	Examples
Budding	Outgrowth on the parent grows and separates	Yeast, hydra
Spore formation	Many light spores formed in a sac, spread by air	Bread mould (*Rhizopus*), *Aspergillus*
Vegetative propagation	New plant from root, stem or leaf	Potato, ginger, Bryophyllum, sugarcane
Artificial methods	Cutting, grafting, layering, tissue culture	Rose, mango, lemon, banana

11.2 Sexual reproduction and meiosis

Sexual reproduction involves **two parents**, and both give genetic material to the child. But here is a problem: if each parent passed on its **full set** of chromosomes, the number would **double every generation** (46 → 92 → 184…). Nature solves this with a special cell division: **meiosis**.

11.2.1 How does meiosis create variation?

- **Chromosomes** are thread-like structures in the nucleus that carry genetic information. Each species has a fixed number.
- Humans have **23 pairs = 46 chromosomes**. In each pair, one came from the mother and one from the father.
- **Meiosis** halves the number: a **diploid** parent cell (2 sets, 46) makes **haploid** daughter cells (1 set, 23). These are the **gametes**.
- Animal gametes: **sperm** (male) and **egg** (female). In plants, the **pollen grain** carries male gametes to the **ovule**, which holds the egg.

- In meiosis, the two chromosomes of each pair **separate**, so every gamete gets **one chromosome from each pair**. Which one goes into a gamete is random.

Mitosis

1 cell → 2 cells. Same chromosome number (46 → 46). Identical cells. For growth, repair and asexual reproduction.

Meiosis

Halves the number (46 → 23). Cells are different from each other. Only for making gametes.

Activity 11.4: the bead model

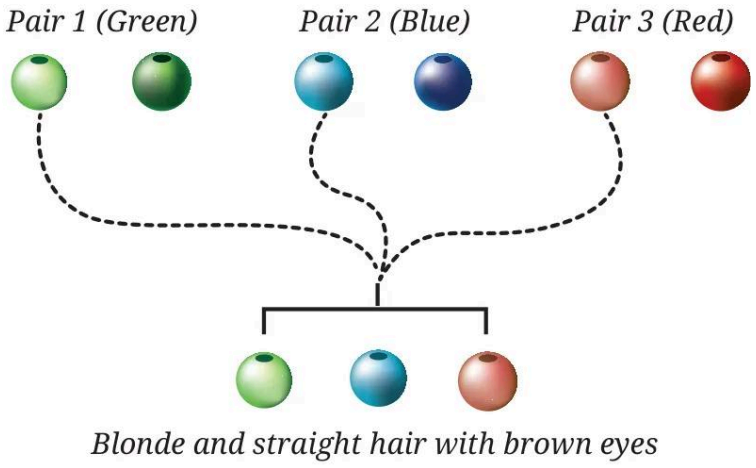


Fig. 11.9 Segregation of characters when gametes form by meiosis

Each pair of beads stands for a pair of chromosomes with two forms of a character:

- **Pair 1 (green)**: light = blonde hair, dark = black hair
- **Pair 2 (blue)**: light = straight hair, dark = curly hair
- **Pair 3 (red)**: light = brown eyes, dark = black eyes

Pick **one bead from each pair** at random. With 3 pairs, there are 2 × 2 × 2 = **8 combinations**.

So why are we all different?

This random mixing gives a huge number of combinations, so children are genetically different from their parents and from their brothers and sisters. This **variation** helps some individuals survive in changing environments, and over time it contributes to **evolution**. Examples: some people can tolerate **low oxygen at high altitudes**; some can **digest milk** as adults.

11.2.2 Sexual reproduction in flowering plants

Flowering plants are called **angiosperms**. They are the most diverse group of plants on Earth, and the **flower is their reproductive organ**. (Non-flowering plants like pines also reproduce sexually; you study them in the next chapter.) Bright colours and fragrance are not just for beauty: they attract pollinators.

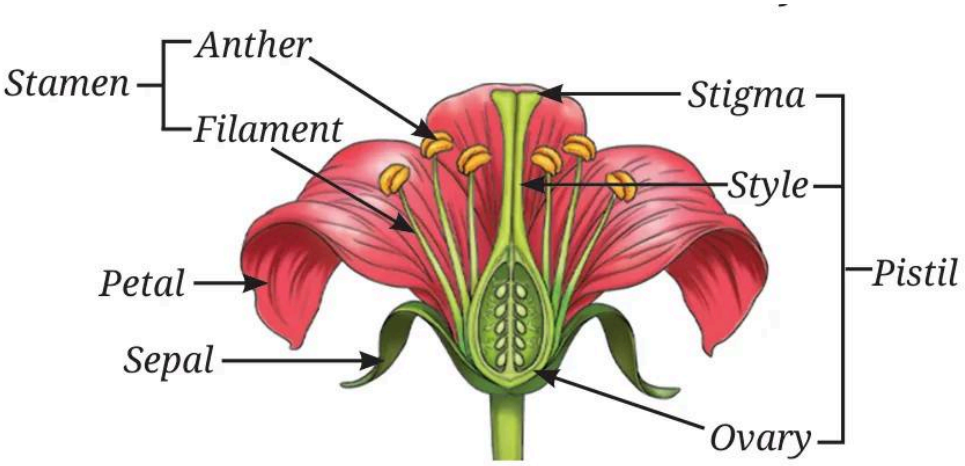


Fig. 11.10 Longitudinal section of a flower

1. **Sepals**: green, outermost whorl. Protect the flower in the bud stage.
2. **Petals**: coloured, often scented. Attract insects and birds.
3. **Stamen** (male): **filament** (stalk) + **anther**, which makes **pollen grains** containing male gametes.
4. **Pistil** (female): **stigma** + **style** + **ovary**.

A **complete flower** has all four parts. In some flowers, sepals and petals are fused together.

The pistil up close

- **Stigma**: at the tip; flat and/or **sticky** to catch pollen.
- **Style**: a thin, long tube joining the stigma to the ovary.
- **Ovary**: the swollen base. It contains **ovules**; each ovule has an **egg cell** (female gamete).

Activity 11.5: cut the ovary **transversely** (T.S.) and **longitudinally** (L.S.) and look under a dissecting microscope: you see the small round **ovules** attached inside.

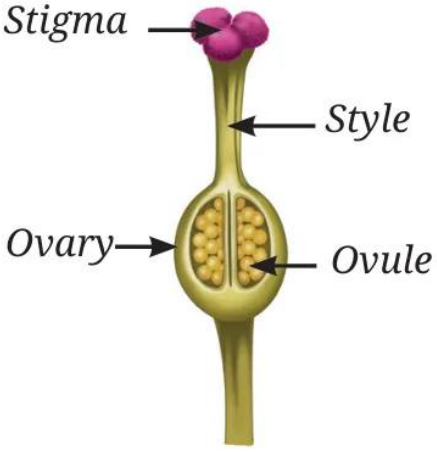


Fig. 11.11 Structure of a pistil

Activity 11.6: is pollen needed for fruits?

On a pea plant, five treatments are set up. Stamens are removed from one bud and one open flower; muslin bags keep insects and outside pollen away.



Fig. 11.12 Experimental set-up for pollination

Treatment	Fruit forms?	Why
Bud, wrapped	**Yes**	Pea pollinates itself inside the bud (self-pollination)
Bud, stamens removed, wrapped	**No**	No pollen of its own, and the bag blocks outside pollen
Open flower, stamens removed, wrapped	**Yes**	It was already pollinated before the stamens were removed
Open flower, wrapped	**Yes**	Own pollen is present
Open flower, not wrapped	**Yes**	Own pollen and pollinators

Conclusion

The transfer of pollen grains from the **anther to the stigma** is necessary for fruit formation. This transfer is called **pollination**.

11.2.3 Pollination

Self-pollination

Pollen reaches the stigma of the **same flower**, or of **another flower on the same plant**. Example: pea, tomato, wheat.

Cross-pollination

Pollen goes from the anther of a flower on **one plant** to the stigma of a flower on **another plant of the same kind**. Example: papaya, sunflower, apple, maize.

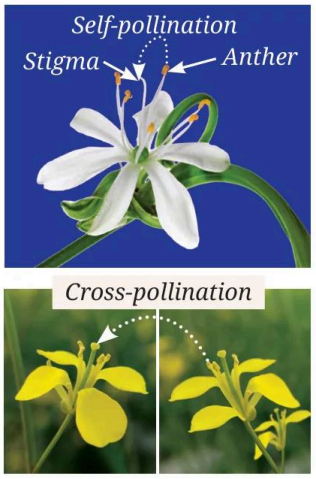


Fig. 11.13 Self and cross-pollination

Point	Self-pollination	Cross-pollination
Pollen from	Same flower or same plant	Another plant of the same species
Needs a pollinator?	Usually not	Yes (wind, water, insects, birds)
Variation	Very little	More: new mixes of characters
Pollen wasted	Less	More

11.2.4 Pollination strategies

Pollination needs outside agents called **pollinators**. Each type of flower is shaped for its pollinator:

🌬️ **Wind** wheat, maize, rice

Pollen is **light, small** and made in **huge numbers**. Stigma is **long and feathery** to trap it. Flowers are dull, no nectar or scent.

💧 **Water** *Vallisneria, Hydrilla*

Water currents carry pollen from flower to flower in aquatic plants.

🐝 **Insects** sunflower, hibiscus, marigold

Flowers are **bright, scented** and give **nectar**. Pollen is **large, sticky or spiny** to cling to the insect. Stigma is sticky.

🐦 **Birds** coral tree, hibiscus

Visited by birds such as the **Indian white-eye** and **sunbirds**, which carry pollen while drinking nectar.

Activity 11.7: which strategy is more efficient?

Strategy	Pollen per flower	Seeds formed
Wind-pollinated grasses (maize, wheat)	5,00,000 – 10,00,000	50 – 200
Insect-pollinated (sunflower)	20,000 – 40,000	800 – 1,000

What the numbers tell us

Wind pollination needs about **2,500 to 20,000 pollen grains for one seed**; insect pollination needs only about **20 to 50**. Insects carry pollen straight to the right flower, so they are far more efficient. Wind is random, so grasses make a huge amount of cheap, tiny pollen. Because grasses grow close together in large fields, enough pollen still lands on stigmas: a big number makes up for the low success rate.

11.2.5

Fertilisation and seed formation

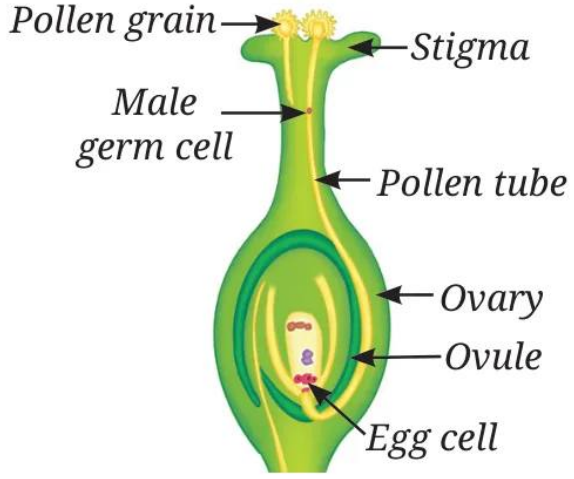


Fig. 11.14 Germination of pollen on stigma

Pollen lands on a compatible stigma

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Pollen grain grows a pollen tube down the style into the ovary

↓

Male gamete travels down the tube to the ovule

↓

Male gamete fuses with the egg cell: fertilisation

↓

Fertilised egg = → grows into an

After fertilisation

Flower part	Becomes
Ovary	Fruit
Ovule	Seed
Zygote	Embryo (inside the seed)
Petals, sepals, stamens	Usually wither and fall



Fig. 11.15 Transformation of an ovary into a fruit (tomato)

Seeds are then **dispersed** by wind, water and animals. When water, air and a suitable temperature are available, a seed **germinates** into a new plant. Sexual reproduction thus makes new plants **and** creates variation, helping species survive and adapt.



Fig. 11.16 (a) Madar seeds: silky hairs, carried by wind



Fig. 11.16 (b) Dandelion seeds: parachutes, carried by wind

Remember the order

Pollination → **G**ermination of pollen → **F**ertilisation → **Z**ygote → **E**mbryo → **S**eed. Say it as: "**P**lease **G**ive **F**resh **Z**ingy **E**xtra **S**eeds".

Meet a scientist: P. Maheshwari



Known as the **"Father of Indian Embryology"** (embryology here is the study of plant reproductive organs). He developed ***in-vitro* fertilisation in flowering plants**, fusing an egg and a male gamete in a test tube to create new hybrid plants, and was one of the first to grow plant embryos on artificial nutrient media. His book *An Introduction to the Embryology of Angiosperms* (1950) is still used worldwide.

Bridging Science and Society: plant breeding

- **Selective breeding**: farmers choose plants with desirable characters to reproduce.
- **Artificial hybridisation**: remove stamens (emasculation), cover the flower with a bag (bagging) to stop self-pollination, then dust pollen with the desired characters by hand.
- **Genetic engineering**: insert genetic material for a desired character into the DNA of a variety.

These give high-yielding and disease-resistant varieties and have revolutionised agriculture.

11.3 Sexual reproduction in animals

Animals reproduce asexually (budding in hydra) and/or sexually. In sexual reproduction, the big challenge is making sure the **sperm meets the egg**, and that the young **survive** long enough to grow up.

Point	External fertilisation	Internal fertilisation
Where	Outside the body, in water	Inside the female's body
How	Female releases eggs into water; male releases sperm over them	Sperm is placed inside the female
Examples	Frogs, most fish	Reptiles, birds, mammals
Eggs made	Very many (hundreds to thousands)	Few
Protection	Little: many eggs are washed away or eaten	Egg/embryo is protected, so survival is higher

11.4 Variation in reproductive strategies

Animal	Habitat	Fertilisation	Eggs at a time	Survival of young

Fish	Water	External	100s – 1000s	Low
Frog	Water/land	External	5,000 – 50,000	Low
Lizard	Land	Internal	2 – 20	Moderate
Bird	Water/land	Internal	1 – 15	Moderate to high

The pattern

Less protection → more eggs. More protection → fewer eggs, but more of them survive.

The yolk problem and the larva

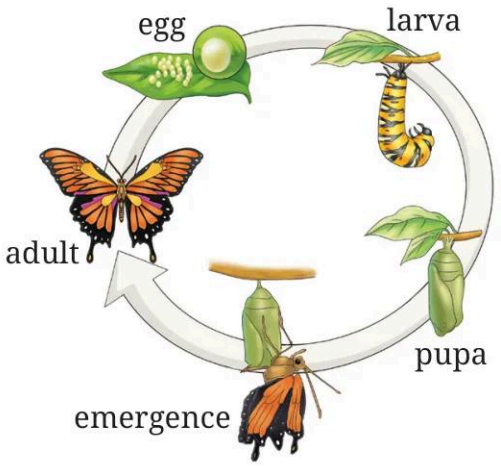


Fig. 11.17 Life cycle of a butterfly

Fish, amphibians and insects lay thousands of eggs, so the mother cannot fill each one with lots of **yolk** (stored food). Each egg has just enough yolk to make a **larva**.

The larva is a **feeding stage**: it eats (rotten food, manure, leaves) and grows. When it has stored enough food, it **transforms** into the adult (butterfly, frog).

Reptiles and birds put **enough yolk** in each egg for the embryo to develop fully before hatching. In mammals, the embryo grows **inside the mother**.

Some young are ready to find food at birth; others need a long period of feeding and care. This depends on how long the embryo develops inside the egg or mother. **Mammals feed their young on breast milk.**

11.5 Reproduction in human beings

11.5.1 Reproductive maturity. As a child grows into an adult (puberty), the reproductive organs **mature** and start producing gametes: **sperm** in males and **eggs** in females. When a sperm meets an egg inside the female body, a **zygote** forms, which develops into an **embryo** and then a **foetus** in the uterus.

11.5.2 Male reproductive system

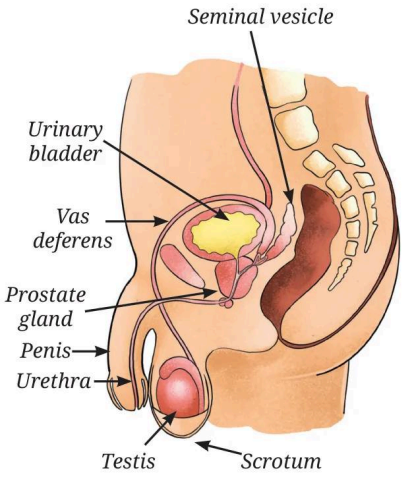


Fig. 11.18 Male reproductive system

1. **Testes** (singular testis): two oval organs that make **sperm** and the male **hormone** that controls sperm production and the changes of puberty in boys.
2. **Scrotum**: skin pouch that keeps the testes slightly **cooler** than body temperature, which sperm formation needs.
3. **Vas deferens**: long tube carrying sperm from the testis to the urethra.
4. **Urethra**: common passage for **urine and sperm**.
5. **Seminal vesicles and prostate gland**: add fluids that **nourish** sperm and help them stay active and move.

Path of sperm

Testis → Vas deferens → Urethra → out through the penis. Think "**T**he **V**ery **U**seful route". A sperm has a **head** (genetic material) and a long **tail** for swimming to the egg.

11.5.3 Female reproductive system

1. **Ovaries** (a pair): make the **eggs** and female **hormones**, which bring about the changes of puberty.
2. **Oviducts / fallopian tubes**: connect each ovary to the uterus. Fertilisation happens here.
3. **Uterus**: bag-like organ where the **foetus develops**.
4. **Cervix**: narrow passage where the uterus opens into the vagina.
5. **Vagina**: passage from the cervix to the outside (birth canal).

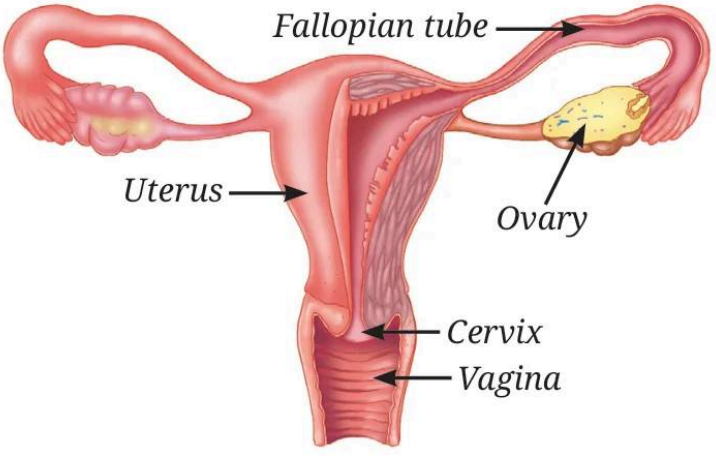


Fig. 11.19 Female reproductive system

Job	Male	Female
Makes gametes and hormones	Testes	Ovaries

Carries gametes	Vas deferens, urethra	Oviducts (fallopian tubes)
Extra parts	Seminal vesicles, prostate (fluids)	Uterus (baby grows), cervix, vagina

11.5.4 Gametes and fertilisation

Making gametes is called **gametogenesis**. It happens in the **testes** and **ovaries** by **meiosis**: body cells have **46** chromosomes, sperm and eggs have **23**. When they join, the zygote again has **46**, the same as the parents.

23 (sperm) + 23 (egg) = 46 (zygote)

Halving in meiosis keeps the chromosome number the same from one generation to the next.

Feature	Sperm	Egg
Size	Very small	Large
Number produced	Millions	Few (usually one a month)
Stored nutrients	Absent	Present
Motility	Actively motile (swims)	Non-motile

This difference between male and female gametes is seen in most animals.

11.5.5 What happens when a sperm meets an egg?

1. At birth, a girl's ovaries already hold **millions of immature eggs**.
2. From puberty, usually **one mature egg** is released each month from one ovary: **ovulation**.
3. Before ovulation, the inner lining of the uterus becomes **thick**.
4. The egg moves from the ovary into the **oviduct**. Sperm that enter the vagina swim up and may meet it there.
5. If a sperm fuses with the egg, a **zygote** forms (fertilisation).
6. The zygote divides by **mitosis** while travelling to the uterus, then **implants** in the thick, blood-rich lining. **Implantation marks the beginning of pregnancy.**

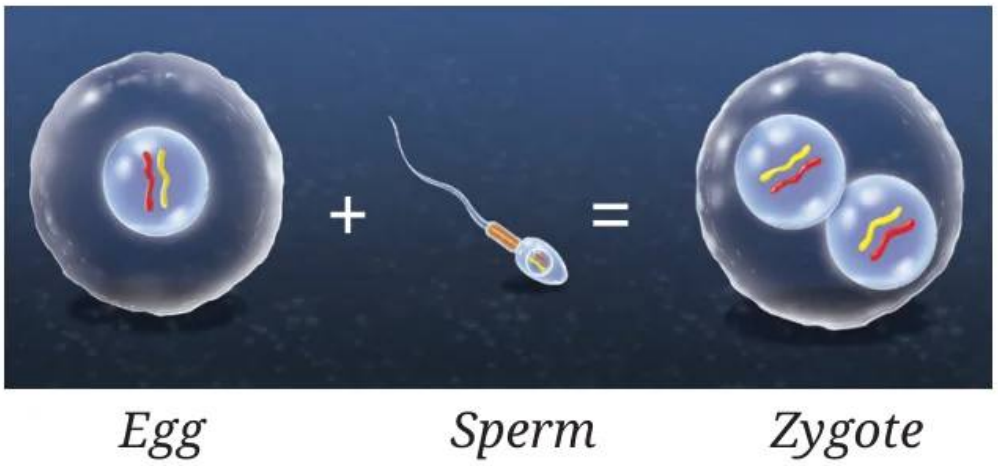


Fig. 11.20 Process of fertilisation

Meet a scientist: Dr Subhash Mukhopadhyay and IVF



In-vitro fertilisation (IVF): an egg and a sperm are joined **outside the body in a laboratory dish**, and the fertilised egg is placed in the uterus to begin a pregnancy. Such a baby is called a "**test tube baby**", though it is really done in a culture dish. In **1978**, **Subhash Mukhopadhyay** of Kolkata made India's first test tube baby, **Kanupriya Agarwal (Durga)**, possible.

11.5.6 The menstrual cycle

If the egg is **not fertilised**, it stays alive for about **a day** and then breaks down. The thick, blood-rich lining of the uterus is no longer needed, so it **sheds** and leaves the body through the vagina with some blood. This is **menstruation** (a period) and lasts about **3 to 7 days**.

The cycle of ovulation, preparing the uterus and menstruation repeats every **21–35 days** (often about **28**). It starts at puberty (age **10–14**) and continues until **menopause** (around age **50**).

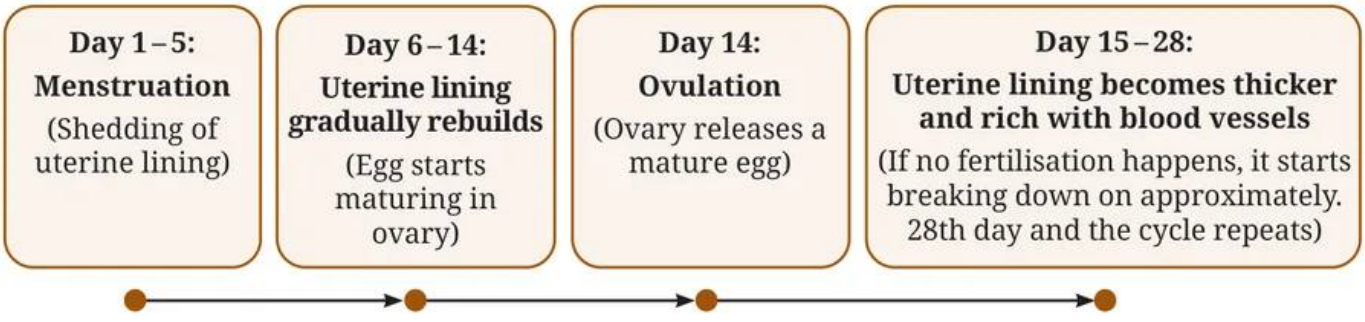


Fig. 11.21 Key stages of the menstrual cycle in a typical 28-day cycle

Is ovulation always on day 14?

No. Day 14 is only for a **typical 28-day** cycle. Ovulation usually happens about **14 days before the next period**, so in a 21-day cycle it is around day 7 and in a 35-day cycle around day 21. It can also shift with stress, illness or diet.

Menstrual hygiene (Bridging Science and Society)

- Use menstrual products: sanitary pads (preferably biodegradable), tampons, menstrual cups.
- Wash the genital area with water regularly; avoid soap there, as it disturbs the natural bacterial balance.
- Always wash hands with soap before and after changing a pad.
- Wrap used pads in newspaper or their wrapper and put them in a bin. Never flush them.
- Wash reusable pads as instructed and dry them completely.
- Change pads every **4–6 hours**, or more often if the flow is heavy.

Period is your pride

Menstruation is a sign of a **healthy** reproductive system, not something to be ashamed of. Clean products, regular changing and responsible disposal keep you healthy and your surroundings clean.

11.5.7

Pregnancy and childbirth

Pregnancy lasts about **nine months** and is divided into three **trimesters** (three months each).

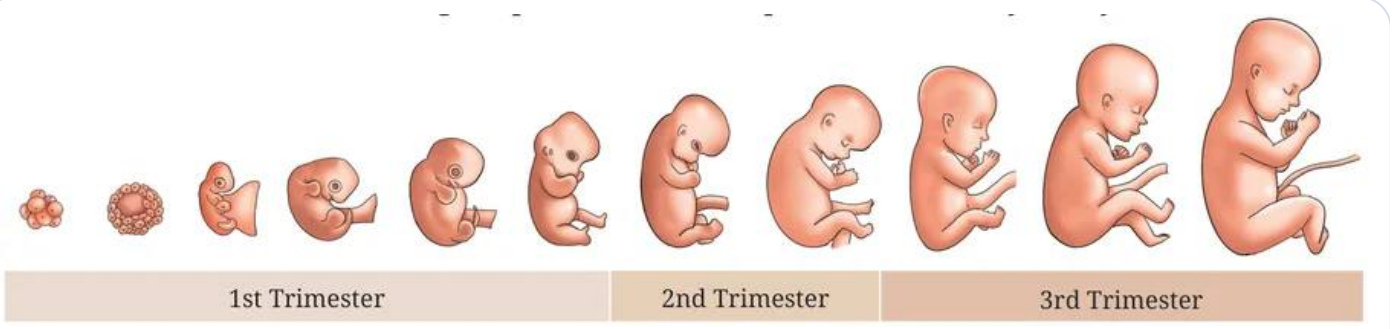


Fig. 11.22 Stages of pregnancy

1st trimester
Zygote → **embryo** in the first two months; major organs start forming. From about the **9th week** it is called a **foetus**.

2nd trimester
Foetus grows bigger and stronger. The mother can usually **feel its movements**.

3rd trimester
Baby grows **rapidly** and gets ready for life outside the womb.

The **uterus** protects and nourishes the baby throughout. At **childbirth**, strong contractions of the uterus muscles push the baby out through the birth canal. If a normal birth is not safe, doctors may use medical or surgical procedures.

Care after birth

- **Breastfeeding**: mother's milk gives complete nutrition and protects the baby from many diseases.
- Keep the newborn **warm**, get **vaccinations on time**, handle gently.
- The mother needs nutritious food, rest, and should avoid smoking, alcohol and medicines without a doctor's advice.

11.5.8 Mother's health during pregnancy

A balanced diet rich in **proteins, vitamins and minerals**, regular **check-ups**, light exercise and rest as advised, and **emotional support** from family keep the mother and baby healthy.

Post-partum depression
Some mothers feel anxiety and tiredness after delivery. It is a recognised, **treatable** condition; they should talk to a doctor, nurse or ASHA worker.

ASHA workers
More than **10 lakh** Accredited Social Health Activists under the **National Health Mission** promote hygiene, immunisation, safe delivery and family planning, especially in villages.

11.5.9 Growing up responsibly

Sexual maturity (sperm production in boys, menstrual cycles in girls) comes gradually in adolescence, but **emotional maturity** takes longer. Being emotionally mature means handling feelings, communicating clearly and making thoughtful decisions. A body that can reproduce does not mean a person is ready for adult responsibilities.

11.5.10 Preventing infections and unwanted pregnancy

Sexually Transmitted Infections (STIs) pass from an infected person through close sexual contact: **gonorrhoea, syphilis, herpes, genital warts, HIV** (which can lead to AIDS). Some are not yet curable. **Condoms** help prevent their spread and also prevent pregnancy.

Method	How it works	Note
Barrier: condom, vaginal cover	Stops sperm reaching the egg	Condoms also protect against STIs
Oral pills	Change hormones so eggs are not released	May have side effects
IUD (Copper-T)	Placed in the uterus to prevent pregnancy	May irritate the uterus
Surgical	Blocks the vas deferens (males) or fallopian tubes (females)	Sperm and egg cannot meet

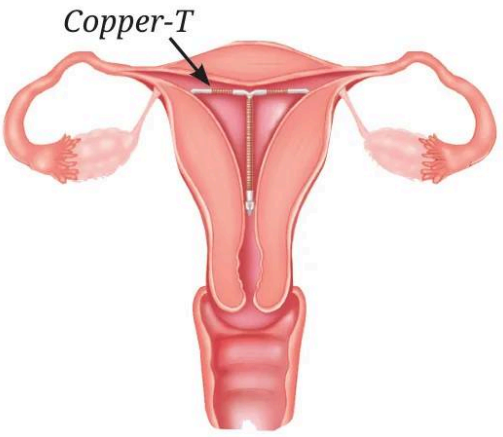


Fig. 11.23 Intra-uterine device (Copper-T)

The law on sex determination

Removing an unwanted pregnancy by surgery is called **abortion** and is generally done only in the first trimester. Choosing abortion because of the baby's sex is a serious concern: it upsets the **sex ratio** of society. That is why **prenatal sex determination is strictly prohibited by law in India.**

Made in India: the weekly pill

Scientists at the **Central Drug Research Institute (CDRI), Lucknow** developed the world's first **non-steroidal, non-hormonal** oral contraceptive pill (centchroman, sold as "Saheli"). Taken **once a week**, it avoids side effects like weight gain, nausea or headaches and is given **free** under the National Family Planning Programme.

Tricks

Memory tricks

Flower parts, outside in S P S P

"**S**ome **P**eople **S**tay **P**retty": **S**epal, **P**etal, **S**tamen, **P**istil.

Which is male? staMEN

Sta**men** has "men" in it, so it is the male part. The pistil is female.

Pistil, top to bottom S S O

Stigma, **S**tyle, **O**vary: "**S**ticky **S**traw to the **O**ven".

What becomes what big → big

The bigger word **ovary** becomes the bigger thing, the **fruit**. The smaller **ovule** becomes the smaller **seed**.

Mitosis vs meiosis copy vs halve

Mitosis = **Mi**rror copy (same 46). **Me**iosis = **Me**et a partner later, so it makes half (23) for gametes.

Menstrual cycle 5 – 14 – 14 – 28

"**Shed, build, release, ready**": days 1–5 shed, 6–14 build, day 14 release the egg, 15–28 ready (then shed again).

Sex of the baby Dad decides

Mother always gives X. **Dad's** sperm brings X (girl) or Y (boy).

Eggs and protection see-saw

More eggs ⇄ less care. Fish and frogs lay thousands; birds lay a few and protect them.

Recap

At a glance

- **Reproduction** produces new individuals of the same species and passes on genetic information.
- **Asexual**: one parent, by **mitosis**, offspring are **clones**. Budding (yeast, hydra), spore formation (bread mould), vegetative propagation (potato, Bryophyllum).
- Artificial vegetative methods: **cutting, grafting, layering, tissue culture**.

- **Sexual**: two parents; **meiosis** halves the chromosomes (46 → 23) to make gametes; random mixing gives **variation**.
- Flower: **sepal, petal, stamen** (anther + filament, male), **pistil** (stigma + style + ovary, female).
- **Pollination**: pollen from anther to stigma; **self** or **cross**; by wind, water, insects or birds.
- Pollen tube carries the male gamete to the ovule → **fertilisation** → zygote → embryo. **Ovule → seed, ovary → fruit.**
- Animals: **external** fertilisation (fish, frog: many eggs, low survival) or **internal** (reptiles, birds, mammals: fewer eggs, better survival).
- Male: testes, scrotum, vas deferens, urethra, seminal vesicles, prostate. Female: ovaries, oviducts, uterus, cervix, vagina.
- **Ovulation** around day 14 of a 28-day cycle; no fertilisation → **menstruation**; fertilisation → implantation → pregnancy (about 9 months, 3 trimesters).
- Birth control: condoms, pills, IUDs (Copper-T), surgery. Condoms also protect from STIs. Prenatal sex determination is illegal in India.

Quiz Test yourself

Tap an answer. **Score: 0 / 12**

1. Tiny plantlets grow on the leaf edges of

Potato	**Bryophyllum**
Ginger	Sugarcane

Bryophyllum leaves sprout plantlets from their notches.

2. In grafting, the rooted plant that receives the stem piece is called the

Scion	**Stock**
Cutting	Layer

Stock = the soil side with roots; scion = the stem piece on top.

3. Bread mould reproduces mainly by

- Budding
- Cutting
- **Spore formation**
- Fertilisation

Spores form in a sac (sporangium) and spread through air.

4. Offspring produced by mitosis are

- **Clones of the parent**
- Haploid
- Very different from the parent
- Gametes

Mitosis makes identical cells with the same chromosome number.

5. How many chromosomes does a human egg cell have?

- 46
- **23**
- 92
- 44

Gametes are haploid: half of 46.

6. With 4 pairs of characters, how many types of gametes are possible?

- 8
- 4
- **16**
- 12

$2 \times 2 \times 2 \times 2 = 2^4 = 16$.

7. The part of the flower that becomes the fruit is the

- Ovule
- **Ovary**
- Stigma
- Anther

Ovary → fruit; ovule → seed.

8. A long, feathery stigma and light pollen suggest pollination by

- Insects
- Birds
- **Wind**
- Water

Wheat, maize and rice are wind-pollinated.

9. Which animal has internal fertilisation?

- Frog
- Fish
- **Bird**
- All of these

Reptiles, birds and mammals: internal.

10. Sperm travel from the testes through the

- Oviduct
- **Vas deferens**
- Cervix
- Scrotum

Testis → vas deferens → urethra.

11. In a 28-day cycle, ovulation usually happens around day

1	5
14	28

Day 14 in a typical 28-day cycle.

12. Which method prevents pregnancy and also protects against STIs?

Oral pill	Copper-T
Condom	Surgery

A barrier stops contact, so it blocks infections too.

Answers to all questions

Tap a question to see the answer. Use **Print or save as PDF** at the top to get all answers open.

Pause and Ponder

1. In a china-rose (hibiscus) plant, a pollen tube grows through the style after pollen lands on the stigma. Which process is about to happen next?

Fertilisation. The male gamete will travel down the pollen tube into the ovule and fuse with the egg cell to form a zygote.

2. Look at the calotropis (madar) and dandelion seeds (Fig. 11.16). What kind of seed dispersal are they adapted for?

Dispersal by wind. Madar seeds have tufts of silky hairs and dandelion seeds have tiny parachutes of hairs. These make them light, so they float far on air currents.

3. A farmer plants two maize varieties side by side; seeds form only when pollen from one variety reaches the stigma of the other. What type of pollination is this?

Cross-pollination: pollen goes from a flower of one plant to the stigma of a flower on another plant of the same species. (In maize it happens by wind.)

4. Why do animals with external fertilisation generally produce more eggs than animals with internal fertilisation?

Eggs laid in water are **unprotected**. Many never meet sperm, and many are washed away by currents or eaten by other animals. Laying thousands of eggs makes sure that at least a few are fertilised and survive. With internal fertilisation, sperm meets the egg inside the body and the embryo is protected, so fewer eggs are enough.

5. In animals, in which fertilisation method are the gametes more protected?

Internal fertilisation. The gametes meet inside the female body, safe from drying, currents and predators, and the embryo also develops in a protected egg or inside the mother.

6. Ravi notices he is growing taller rapidly, his shoulders are broadening and his voice cracks. What stage of life is he entering?

Puberty (the start of adolescence). The testes have started producing the male hormone, which brings these changes and marks reproductive maturity.

7. Rina's period occurs every 28 days. Her last period was on 5th March. When is she most likely to get her next period?

5 March + 28 days. March has 31 days, so 5 + 28 = 33 → 33 − 31 = **2nd April**. (Try it in the calculator in Section 11.5.6.)

8. A human zygote has just formed. How many chromosomes does it have?

46 (23 pairs): 23 from the sperm + 23 from the egg.

9. What protective devices can be used during sexual activity to reduce the spread of STIs?

Condoms (barrier methods). They stop direct contact with body fluids, so they reduce the spread of STIs such as HIV, gonorrhoea and syphilis, and also prevent pregnancy.

10. If a couple uses oral contraceptive pills but not condoms, which risks remain and why?

The risk of **sexually transmitted infections** remains. Pills only change hormones so that eggs are not released; they do **not** block contact between partners, so infections can still pass. (Pills can also fail if a dose is missed.)

11. Young ones of many animals can walk or find food soon after birth, but human babies are dependent for a long time. Advantages and disadvantages for humans as a species?

Advantages: the brain keeps developing after birth, so children have a long time to **learn** language, skills and culture from parents; strong family bonds and care.
Disadvantages: babies are **helpless and vulnerable** for years; parents must spend a lot of time, energy and resources; humans have few children, so the population grows slowly.

Activity 11.7

1. Compare the two strategies in terms of pollen-to-seed ratio and efficiency of pollination and seed formation.

Wind: 5,00,000 ÷ 200 = 2,500 up to 10,00,000 ÷ 50 = 20,000 pollen grains per seed.
Insect: 20,000 ÷ 1,000 = 20 up to 40,000 ÷ 800 = 50 pollen grains per seed.
Insect pollination is roughly **100 to 1,000 times more efficient**: it makes much less pollen but forms many more seeds. Most wind-blown pollen is wasted.

2. Why can producing a very large number of pollen grains still be an effective strategy?

Wind is random and free. Wind-pollinated pollen is tiny and cheap to make, so the plant can make lakhs of grains. Grasses like wheat and maize grow close together in large numbers, so with so much pollen in the air, enough grains land on the long, feathery stigmas to form seeds. The plant also saves energy by not making big petals, scent or nectar.

Revise, Reflect, Refine

1. **Anthers are removed before the flower matures; later pollen from another plant of the same species is dusted on its stigma and seeds form. Which process has been ensured? (i) Self-pollination (ii) Cross-pollination (iii) Fertilisation (iv) Tissue culture**

(ii) Cross-pollination. Removing the anthers prevents self-pollination, and the pollen came from another plant. (Fertilisation followed, but the step deliberately ensured is cross-pollination, as in artificial hybridisation.)

2. **Arrange in order: (i) Pollen germination on stigma (ii) Fertilisation (iii) Pollination (iv) Formation of zygote**

(iii) → (i) → (ii) → (iv): Pollination → pollen germination on stigma → fertilisation → formation of zygote.

3. **Assertion (A): The zygote formed after fertilisation immediately attaches to the uterus wall. Reason (R): The uterus wall is always prepared to receive the zygote.**

A is false. The zygote forms in the oviduct and divides by mitosis while it travels to the uterus; only then does it implant.
R is not fully correct either: the lining is not always ready; it thickens and becomes blood-rich in each cycle (after ovulation) and is shed during menstruation.
Among the options, the expected answer is **(iv) A is false, but R is true**, reading R as "the uterus wall prepares itself to receive the zygote". In your answer, mention that the preparation happens every cycle, not all the time.

4. **Why does asexual reproduction produce offspring that are genetically identical to the parent?**

Only **one parent** is involved and there is no fusion of gametes. New cells are made by **mitosis**, which copies the chromosomes exactly and gives each daughter cell the same number and the same genetic information as the parent cell. So the offspring are clones.

5. Explain why the menstrual cycle stops during pregnancy.

Menstruation is the shedding of the uterine lining when no embryo has implanted. In pregnancy, the embryo is **implanted** in that thick, blood-rich lining and depends on it for nourishment, so the lining must be **kept**, not shed. Hormones during pregnancy also stop new eggs from maturing and being released, so ovulation and menstruation stop until after childbirth.

6. Why are flowers that bloom at night white or light coloured, compared to day-blooming flowers?

Night flowers are pollinated by night-active animals such as **moths and bats**. Bright colours cannot be seen in the dark, but **white or pale petals reflect** the little moonlight and are easy to spot. Such flowers are usually also **strongly scented** (e.g. raat ki rani, jasmine) to guide pollinators by smell. Day flowers use bright colours to attract bees, butterflies and birds.

7. Why are vegetatively propagated plants more vulnerable to diseases than sexually reproduced plants?

All vegetatively propagated plants are **genetically identical clones**. If a disease can attack one of them, it can attack all of them, and none has a variation that gives resistance, so a whole crop can be wiped out. Diseases (especially viruses) in the parent are also passed directly to the new plants. Sexually produced plants show **variation**, so some may resist the disease and survive.

8. If all flowers of a plant type could only self-pollinate, how would genetic diversity change over generations?

Genetic diversity would **fall**. The genetic material would only be reshuffled within the same plant, so no new characters would come in from other plants. Over generations the plants would become more and more alike (uniform). Such a population would adapt poorly to new diseases, pests or climate change, and harmful characters could build up.

9. A farmer wants many genetically identical plants quickly. Suggest suitable methods and why they work.

Tissue culture (fastest: thousands of disease-free plantlets from a few cells, e.g. banana), and **cutting, grafting and layering** (sugarcane, rose, mango, lemon). These are asexual methods based on mitosis, so every plant is an exact copy of the chosen parent with the same good qualities. They skip seed formation and flowering, so plants are ready sooner, and the fruit quality is predictable.

10. Suresh tests pollen germination in sugar solutions of 0%, 2.5%, 5%, 7.5% and 10%. (i) What hypotheses can be tested? (ii) What should be kept the same?

(i) Hypotheses:

- Pollen grains need sugar to germinate (compare 0% with the others).
- Germination increases with sugar concentration up to a certain level.
- Too much sugar reduces germination (high concentration draws water out of the pollen).
- There is an optimum sugar concentration at which most pollen germinates and pollen tubes grow longest.

(ii) Keep the same: pollen from the same species and same flower age, number/amount of pollen, volume of solution, temperature, light, time of observation, type of slide and moisture, and the method of counting germinated grains. Only the sugar concentration should change.

11. Which type(s) of pollination might have been followed in tomato (stamens cover the stigma), wheat (flowers open after pollination) and papaya (male and female flowers on different trees)?

Tomato	Wheat	Papaya
		
Stamens cover the stigma.	Flowers open after pollination.	Male and female flowers are often borne on different papaya trees.

Pictures from Question 11

- **Tomato: self-pollination.** The stamens form a cone around the stigma, so its own pollen falls on it.
- **Wheat: self-pollination.** Pollination happens before the flower opens, so outside pollen cannot enter.
- **Papaya: cross-pollination.** Male and female flowers are on different trees, so pollen must travel from a male tree to a female tree (by wind and insects).

12. Apple orchards: Place A with natural pollinators, Place B with bee colonies (Fig. 11.24). (i) Hypotheses? (ii) Parameters? (iii) Compare A and B. (iv) Inference?

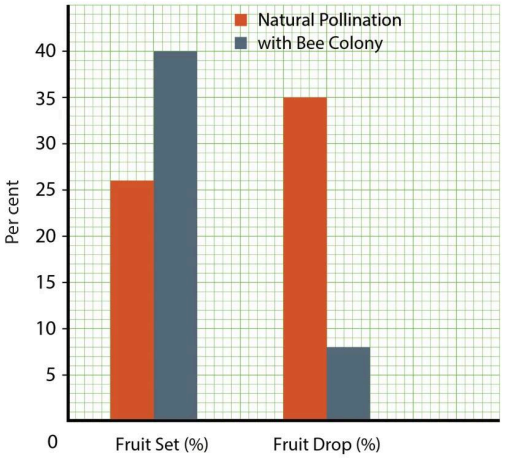


Fig. 11.24 Fruit set and fruit drop in the two orchards

(i) Hypotheses: keeping bee colonies in orchards increases pollination and therefore fruit set; better pollination reduces premature fruit drop; the fall in apple yield is due to a decline in natural pollinators.

(ii) Parameters: *variable changed*: type of pollination (natural vs bee colony). *Measured*: fruit set (%) and fruit drop (%). *Kept similar*: apple variety, age of trees, number of branches, climate, soil, irrigation, fertiliser and pest control, season.

(iii) Comparison:

	Place A (natural)	Place B (bees)
Fruit set	26%	40%
Fruit drop	35%	8%

Place B has about 1.5 times the fruit set and less than a quarter of the fruit drop, so it gives a much higher final yield.

(iv) Inference: bees improve pollination; well-pollinated flowers form more fruits that stay on the tree. Beekeeping with apple farming (mixed farming) increases apple yield and also gives honey, so it is a good way to fight the loss of natural pollinators.

13. A student claims, "In humans, ovulation always happens on day 14 of the menstrual cycle." Is this correct? Give at least two reasons.

The claim is **not correct**.

1. Cycle length differs from person to person (21–35 days). Day 14 fits only a 28-day cycle.
2. Ovulation happens about 14 days **before the next period**, so in a 21-day cycle it is near day 7 and in a 35-day cycle near day 21.
3. Even in the same person, it can shift from month to month because of stress, illness, diet, travel or hormonal changes, and is often irregular just after puberty.

Threads of Curiosity and The Quest Continues

- **Who determines the sex of a baby?**

The **father**. The mother's egg always carries X. A sperm carries either X (baby XX, girl) or Y (baby XY, boy). So the sperm that fertilises the egg decides.

- **Does a unicellular organism like amoeba or yeast ever "grow old"?**

When an amoeba divides into two, the parent does not die and leave a body behind; it simply becomes two new young cells. In that sense it does not age like we do, and its line can go on and on. But scientists have found that ageing still happens in some ways: in **yeast**, which reproduces by budding, the mother cell keeps a scar for each bud, collects damage and can make only a limited number of buds before it dies, while the buds start "young". So ageing exists, but it looks very different from ageing in humans.

The Journey Beyond: cutting a tomato



(a) L.S. of tomato fruit



(b) T.S. of tomato fruit



(c) L.S. of tomato flower

The chambers and seed attachment you see in the tomato fruit match the inside of the **ovary** of the tomato flower, because the fruit **is** the ripened ovary and the seeds are the ripened ovules.

Diagrams from the NCERT Exploration textbook (Grade 9).